

Conservation of Energy

$$(KE + PE + E_{\text{non conservative force}})_{\text{initial}} = (KE + PE)_{\text{Final}}$$

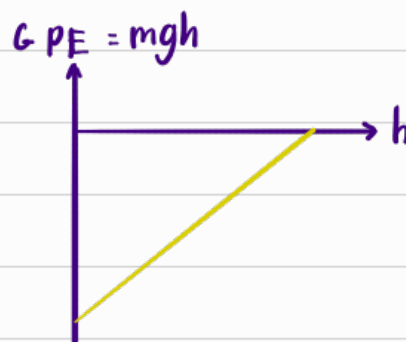
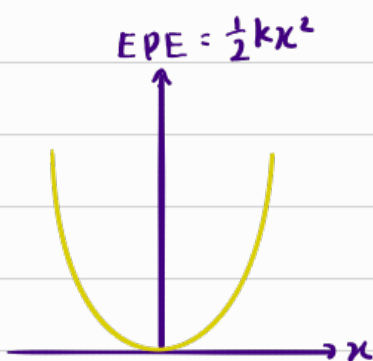
Relationship btwn PE & Force

$$F = - \frac{dU}{dx}$$

Naturally occurring forces will always want an object to have the lowest possible energy state

↳ Bcz the less energy, the more stable

- Example :
- Spring get extended by external force to the right
 - $EPE \uparrow \Rightarrow \frac{dU}{dx} = +ve$
 - Spring force will go to the left to cause EPE to $\downarrow$
 - Thus the negative so that the force always opposes the increase in potential energy



• Since gradient is always positive, F_g is always negative.

• That's why F_g is always downwards

Turning Point

Not the Math turning point, but the literal meaning

- The point which an object is gonna change direction (turn)
- This happens when all $KE \rightarrow PE$, meaning $PE = \text{Total Mechanical Energy}$.

